

Chatting Platform using HashMap and Queue

Project Overview:

This project implements a simplified chatting platform that allows multiple users to communicate by sending and receiving messages.

The system focuses on applying **Data Structures concepts** in a real-life scenario.

- **HashMap** is used to manage users efficiently and allow fast access using usernames.
- **Queue** is used to store and deliver messages in the correct order (FIFO).

Each user has a personal inbox where messages are stored and retrieved sequentially.

Project Objectives

- Apply **HashMap** for fast user lookup and session management.
 - Apply **Queue** to preserve message order.
 - Simulate real chat behavior (send, receive, delete users).
 - Separate system logic from user interface (Console / GUI).
 - Demonstrate understanding of core data structures.
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Data Structures Used

Queue (InboxQueue)

- Used to store incoming messages for each user.
- Follows **FIFO (First In First Out)**.
- Messages are:
 - Added at the **rear**
 - Removed from the **front**

Why Queue?

Because chat messages must be delivered in the same order they were sent.

HashMap

- Used to store and manage users.
- Maps:
 - **Key** → Username
 - **Value** → Index of the user in the users array

Why HashMap?

It provides fast access to users in **O(1)** average time, instead of searching linearly.

System Architecture (High Level)

Components:

- **User Module:** Stores user data and inbox queue.
 - **Queue Module:** Manages message ordering.
 - **HashMap Module:** Handles user lookup.
 - **ChatSystem Module:** Core logic (send, receive, login, register).
 - **Interface Module:** Console / GUI for user interaction.
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Message Flow (How the system works)

1. User logs in using username.
2. HashMap searches for the user using the username as a key.
3. If found, user gains access to the system.
4. When sending a message:
 - Receiver is located using HashMap.
 - Message is added to receiver's inbox queue (enqueue).
5. When reading messages:
 - Messages are removed from inbox queue (dequeue) in order.

Test Cases

Chat System 2.0

Welcome to Chat System

Please select an option to continue

Login

Signup

Exit

Create Account

Join us today!

Username:malak

Phone:010

Password:malak123

Sign Up

Back

Exit App

Dashboard - malak

Send messages or check your inbox

To (User):

Message:

Send Message

Check Inbox

Logout

Exit App

Dashboard - malak

Send messages or check your inbox

To (User):

Message:

Send Message

Check Inbox

Logout

Exit App

